Likewise, when this correct symmetry is present in the frequency domain, the Inverse DFT will produce a time domain that has an imaginary part composes of all zeros (plus round-off noise). These debugging techniques are essential for using the FFT; become familiar with them.

Speed and Precision Comparisons

When the DFT is calculated by correlation (as in Table 12-2), the program uses two nested loops, each running through *N* points. This means that the total number of operations is proportional to *N times N*. The time to complete the program is thus given by:

EQUATION 12-1
DFT execution time. The time required to calculate a DFT by correlation is proportional to the length of the DFT squared.

$$ExecutionTime = k_{DFT} N^2$$

where N is the number of points in the DFT and k_{DFT} is a constant of proportionality. If the sine and cosine values are calculated *within* the nested loops, k_{DFT} is equal to about 25 microseconds on a Pentium at 100 MHz. If you *precalculate* the sine and cosine values and store them in a look-up-table, k_{DFT} drops to about 7 microseconds. For example, a 1024 point DFT will require about 25 seconds, or nearly 25 milliseconds per point. That's slow!

Using this same strategy we can derive the execution time for the FFT. The time required for the bit reversal is negligible. In each of the $Log_2 N$ stages there are $N/2$ butterfly computations. This means the execution time for the program is approximated by:

EQUATION 12-2
FFT execution time. The time required to calculate a DFT using the FFT is proportional to *N* multiplied by the logarithm of *N*.

$$ExecutionTime = k_{FFT} N \log_2 N$$

The value of k_{FFT} is about 10 microseconds on a 100 MHz Pentium system. A 1024 point FFT requires about 70 milliseconds to execute, or 70 microseconds per point. This is more than 300 times faster than the DFT calculated by correlation!

Not only is $NLog_2 N$ less than N^2, it increases much more slowly as N becomes larger. For example, a 32 point FFT is about *ten* times faster than the correlation method. However, a 4096 point FFT is *one-thousand* times faster. For small values of N (say, 32 to 128), the FFT is important. For large values of N (1024 and above), the FFT is absolutely critical. Figure 12-8 compares the execution times of the two algorithms in a graphical form.